

LIONAEL SURYA DWITAMA

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PROFESSIONAL SUMMARY

Honours Bachelor of Data Science and Analytics student (3.9 GPA, President's Honour List) with hands-on experience building and validating quantitative models end to end in Python — from statistical feature design through calibrated machine-learning models to production-facing tools. Comfortable translating rule-heavy, regulation-driven logic into tested code, benchmarking competing modeling approaches under rigorous validation, and communicating model behavior to non-technical stakeholders. Seeking to bring this quantitative and programming foundation to the Derivative and Asset Modeling team's model development, validation, and market data research workstreams.

EDUCATION

Seneca Polytechnic — Honours Bachelor of Data Science and Analytics

Expected Apr 2028

Newnham Campus, Toronto | Cumulative GPA: 3.9/4.0 | President's Honour List (Winter 2025, Winter 2026)

- Relevant coursework: Simulation and Modelling (A+), High Performance Computing (A), Statistical Methods for Data Science (A+), Math Methods for Data Science, Data Mining (A+), Data Preparation (A+), Programming for Data Science (A+), Database Systems, Business Intelligence.

TECHNICAL SKILLS

Programming: Python (primary), SQL, JavaScript/TypeScript

Quantitative & Statistical Methods: statistical modeling, hypothesis testing, PCA, time-series forecasting, walk-forward validation, feature engineering, probability calibration

Modeling & ML: XGBoost, SHAP, model calibration, threshold optimization, Prophet

Data Engineering & Tools: Kafka, Spark Structured Streaming, PostgreSQL, Power BI, Excel, Git/GitHub, FastAPI

QUANTITATIVE & MODELING PROJECTS

IDX Stock Signal Scanner

Personal project — Python, XGBoost, Claude API

- Built a three-stage equity screening pipeline for the Indonesia Stock Exchange: a rules-based filter to narrow the investable universe, an XGBoost model to rank candidate signals, and an LLM explanation layer to generate the rationale behind each ranked signal.
- Designed the pipeline so quantitative ranking and human-readable justification are produced together, making model output auditable rather than a black box — a workflow directly analogous to reviewing and explaining model-driven valuation output.
- Implemented the full pipeline in Python: data ingestion, feature construction, model scoring, and explanation generation.

Retail Demand Forecasting — Model Benchmarking Study

Personal project — Python, Prophet, XGBoost

- Built and benchmarked competing forecasting approaches — a statistical decomposition model (Prophet) and a gradient-boosted tree model (XGBoost) — against the same retail demand series.
- Used walk-forward validation to evaluate each model only on data that would have been available at the time of prediction, avoiding lookahead bias and mirroring how models are monitored and reassessed over time in production.
- Quantified the accuracy/interpretability trade-off between the two modeling families to inform which approach was more appropriate under different data conditions.

Customer Churn Prediction — ROI-Optimized Classifier

Personal project — Python, XGBoost, SHAP, FastAPI, Streamlit

- Developed a calibrated XGBoost classifier and explicitly calibrated its output probabilities so the predicted risk scores could be trusted as decision inputs rather than raw scores.
- Set the classification threshold by optimizing expected ROI — explicitly modeling the asymmetric cost of false positives versus false negatives — instead of defaulting to an accuracy-maximizing cutoff.
- Used SHAP to explain individual predictions and shipped the model via a FastAPI service with a Streamlit interface and drift monitoring to track performance degradation post-deployment.

EXPERIENCE

Software Engineering Intern

Jun 2025 – Aug 2025

CV. Veda Sakti Dharma | Jakarta, Indonesia

- Built a rules-based calculation engine that translated Indonesia's PPh 21 payroll tax regulations into tested, automated Python logic — experience directly relevant to encoding regulation- and formula-driven financial logic into reliable code.
- Validated payroll data and debugged calculation discrepancies across edge cases, and collaborated with the engineering team to implement new features through the development cycle.

Data Entry and Web Development

Jul 2025 – Present

QuickRn Canada

- Investigated and resolved data/access issues through technical troubleshooting for non-technical stakeholders.